

Electricity Prices - Process Connection

This process connection retrieves day-ahead electricity market prices for **today** and **tomorrow** and provides them hour by hour, as well as daily figures (minimum, maximum, average, current price, cheapest time window) in the visualization. This lets the customer shift consumption into cheap hours.

Supported Countries

Country	Source	Resolution	License
DE, AT	aWATTar API	full daily series	free, fair use 100 requests/day
FR, NL	Energy-Charts.info (Fraunhofer ISE)	full daily series	CC BY 4.0 - attribution required
KR	Elecz.com (KPX SMP)	current price only, no daily series	free, commercial use allowed

Not supported: China (no free/public day-ahead API found); Spain and Lithuania (technically available via Energy-Charts.info, but their license only allows private/internal use - a customer-facing display is therefore not covered).

Configuration

These values are set by the installer in the **Configuration** folder:

Field	Meaning
Country (COUNTRY)	DE, AT, FR, NL or KR (empty = DE). Unknown codes safely fall back to DE.
Surcharge (SURCHARGE_CT)	Grid fees/levies/margin in ct/kWh, added to the market price. DE/AT/FR/NL only.
VAT (VAT_PERCENT)	VAT rate in %, applied to market price + surcharge. DE/AT/FR/NL only.
Poll interval (POLL_MINUTES)	Interval for price retrieval in minutes (default/minimum 15).
Cheap threshold (THRESHOLD_CT)	Price in ct/kWh below which 'currently cheap' switches to yes. DE/AT/FR/NL only.
Cheapest time window (WINDOW_HOURS)	Length of the cheapest contiguous time window to search for, in hours (default 4).

Status

Field	Meaning
Error status / error text	Shows whether the last price retrieval failed, and why.
Last update	Timestamp of the last successful price retrieval.
Refresh now	Button to trigger an immediate price retrieval.
Source resolution	Time resolution delivered by the source in minutes (e.g. 60 or 15); empty for KR.
Price unit	Currency/unit of the currently active source: ct/kWh (DE/AT/FR/NL) or KRW/kWh (KR).

Prices Today / Tomorrow

The **Prices today** and **Prices tomorrow** folders contain the same fields: date, availability, daily minimum/maximum/average, hour of minimum/maximum, start hour and average price of the cheapest time window, and 24 hourly prices (00:00-01:00 through 23:00-00:00). Only the 'Prices today' folder additionally has: current price, current hour, and the 'currently cheap' flag.

For DE/AT/FR/NL all prices are consumer prices: $(\text{market price[EUR/MWh]} / 10 + \text{surcharge}) * (1 + \text{VAT}/100)$.
For KR, the current price is the unmodified KPX wholesale price in KRW/kWh, without surcharge/VAT and without a daily series - 'Prices tomorrow' stays empty for KR.

Usage Notes

- Prices for 'tomorrow' are usually published by the exchange only from around 2 pm.
- If a price retrieval fails, the last known values are kept; only the error status is set.
- On days with a daylight-saving time change, missing hours are shown empty instead of a stale value.

License Notice

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